MTH2008

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

January 2024

Real Analysis

Module Leader: Dr Julian Newman

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) Give the definition of a boundary point of a set $S \subset \mathbb{R}$. (5)

   (b) State whether each of the following sets is open, closed, both or neither. (You are not required to write a justification.)

   (i) $\emptyset$
   (ii) $\{0\} \cup [1, 3)$
   (iii) $\{0\} \cup [1, \infty)$
   (iv) $\{\frac{1}{n} \mid n \in \mathbb{Z} \setminus \{0\}\}$
   (v) $\bigcup_{n=1}^{\infty} [\frac{1}{2n+1}, \frac{1}{2n}]$ (5)

   (c) Provide a proof or counterexample to each of the following statements:

   (i) Every subset of $\mathbb{N}$ has an infimum.
   (ii) For every non-empty bounded subset S of $\mathbb{R}$,

   $$(\sup S)^2 = \sup \{x^2 \mid x \in S\}.$$

   (iii) For every non-empty bounded subset S of $\mathbb{R}$, the set

   $$\{x^2 \mid x \in S\}$$

   has a supremum. (15)

   (d) Let us define the function

   $$f \colon \mathbb{R} \to \mathbb{R}$$
   $$f(x) = \begin{cases} \frac{x}{1+|x|} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0. \end{cases}$$

   (i) Show, from the ε-δ definition of limits, that $f(x) \to 0$ as $x \to 0$.
   (ii) Determine whether f is continuous at 0. Justify your answer. (10)

   (e) Let us define the function

   $$g \colon [-2, 2] \to \mathbb{R}$$
   $$g(x) = \begin{cases} 0 & \text{if } x \in [-2, 2] \setminus \{0\} \\ 1 & \text{if } x = 0. \end{cases}$$

   For each $n \in \mathbb{N}$, let us define the partition P_n of $[-2, 2]$ by

   $$P_n = (-2, -\tfrac{1}{n}, \tfrac{1}{n}, 2).$$

(i) Find the value of the upper Riemann sum $U(g, P_n)$ and the lower Riemann sum $L(g, P_n)$ for each $n \in \mathbb{N}$.

(ii) Show, from the definition of Riemann integrals, that g is Riemann-integrable with

$$\int_{-2}^{2} g(x)\,\mathrm{d}x = 0.$$

(15)

[**50**]

SECTION B

2. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of real numbers.

   (a) Define what it means for the sequence $(a_n)_{n\in\mathbb{N}}$ to be Cauchy. (5)

   (b) Using Cauchy's convergence criterion, show that if $\sum_{n=1}^{\infty} |a_n|$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent. (10)

   (c) Using Cauchy's convergence criterion, show that if $(a_n)_{n\in\mathbb{N}}$ is bounded and

   $$|a_{n+1} - a_n| \leq \frac{|a_{n+2}|}{2^n}$$

   for all $n \in \mathbb{N}$, then $(a_n)_{n\in\mathbb{N}}$ is convergent. (10)

   [25]

3. (a) (i) Let us define a sequence $(f_n)_{n\in\mathbb{N}}$ of functions $f_n\colon \mathbb{R}\to\mathbb{R}$ by

$$f_n(x)=\begin{cases} x^2+1 & \text{if } |x|\le n\\ 0 & \text{if } |x|>n\end{cases}$$

for every $n\in\mathbb{N}$. Show that (f_n) is pointwise convergent on $\mathbb{R}$, and determine whether (f_n) is uniformly convergent on $\mathbb{R}$.

(ii) Let us define a sequence $(g_n)_{n\in\mathbb{N}}$ of functions $g_n\colon \mathbb{R}\to\mathbb{R}$ by

$$g_n(x)=\begin{cases} \frac{1}{x^2+1} & \text{if } |x|\le n\\ 0 & \text{if } |x|>n\end{cases}$$

for every $n\in\mathbb{N}$. Show that (g_n) is pointwise convergent on $\mathbb{R}$, and determine whether (g_n) is uniformly convergent on $\mathbb{R}$. (18)

(b) *Recall that for any real number $y\in\mathbb{R}$, the number $\lceil y\rceil\in\mathbb{Z}$ is the integer obtained by rounding y up; in other words,*

$$\lceil y\rceil=\min\{n\in\mathbb{Z}\,|\,n\ge y\}.$$

Let us define the function

$$h\colon [0,1]\to\mathbb{R}$$
$$h(x)=\begin{cases}\frac{1}{\lceil\frac{1}{x}\rceil} & \text{if } x\in(0,1]\\ 0 & \text{if } x=0.\end{cases}$$

Show that h is Riemann-integrable on $[0,1]$. (7)

[**25**]

4. (a) Let us define a function $f\colon \mathbb{R}^3 \to \mathbb{R}$ by

$$f(x_1, x_2, x_3) = x_1 x_2 x_3 \quad \forall\, (x_1, x_2, x_3) \in \mathbb{R}^3.$$

Show that f is differentiable at every point $(x_1, x_2, x_3) \in \mathbb{R}^3$, and find the Jacobian $Df(x_1, x_2, x_3)$. (5)

(b) Let us define a function $g\colon \mathbb{R}^2 \to \mathbb{R}$ by

$$g(x_1, x_2) = \begin{cases} \frac{x_1^2 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq (0, 0) \\ 0 & \text{if } (x_1, x_2) = (0, 0). \end{cases}$$

(i) Show that the Jacobian Dg exists at $(0, 0)$.

(ii) Show that g is not differentiable at $(0, 0)$. (15)

(c) Suppose we have functions $F\colon \mathbb{R} \to \mathbb{R}$ and $G\colon \mathbb{R} \to \mathbb{R}$ that are both differentiable on $\mathbb{R}$. Suppose the derivative F' of F is given by

$$F'(x) = G(F(x)) \quad \forall\, x \in \mathbb{R}.$$

Show that F is twice differentiable on $\mathbb{R}$. (5)

[**25**]